



Senior Design Project Mid-Term Evaluation Presentation

on

Web Based Diabetes Prediction System Using Machine Learning

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Presentation Outline

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Introduction

■ Overview/Specifications

- This project proposes a machine learning-based web application to predict diabetes using patient data.
- Analyzes factors like glucose levels, BMI, age, blood pressure, Insulin, skin thickness ,diabetes pedigree function, and insulin to predict diabetes.
- The goal is to improve prediction accuracy beyond traditional methods and make the process faster and accessible encouraging at-risk individuals to adopt healthier lifestyle changes.



Motivation

- Indians are prone to diabetes due to a combination of lifestyle and genetic factors, with over 100 million people currently affected. As this trend continues to rise, an early detection system would be highly beneficial in identifying at-risk individuals and alerting them in time for preventive measures and timely intervention.
- In current medical diagnosis method, there can be three different types of errors
 - The false-negative type in which a patient in reality is already a diabetic patient but test results tell that the person is not having diabetes.
 - The false-positive type. In this type, patient in reality is not a diabetic patient but test reports say that he/she is a diabetic patient.
 - The third type is unclassifiable type in which a system cannot diagnose a given case. This happens due to insufficient knowledge extraction from past data, a given patient may get predicted in an unclassified type.
- However, in reality, the patient must predict either to be in diabetic category or non-diabetic category. Such errors in diagnosis may lead to unnecessary treatments or no treatments at all when required. In order to avoid or reduce severity of such impact, there is a need to create a system using machine learning algorithm and data mining techniques which will provide accurate results and reduce human efforts

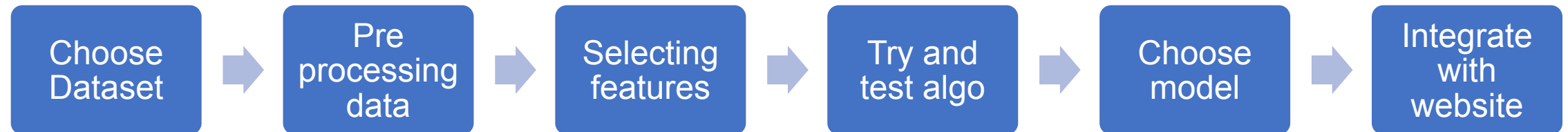


Literature Review /Existing or Similar Product/Process/Algorithm

- The analysis o related or gives results on various healthcare datasets here analysis and predictions ere carried out using various methods and techniques. Various prediction models have been developed and implemented by various researchers using variants o data mining techniques, machine learning algorithms or also combination of these techniques. Dr Vaidehi V founds 98.8% using Ada Boost classifier
- <https://www.sciencedirect.com/science/article/pii/S1877050920300557>
- Hasan, M. K., Alam, M. A., Das, D., Hossain, E., et al. (2020). Diabetes prediction using ensembling of different machine learning classifiers. IEEE Access. Retrieved from ieeexplore.ieee.org. Finds 94% accuracy using Adaboost and 94.6% using XG Boost
- <https://ieeexplore.ieee.org/document/9076634?denied=>
- Sonar, P., & JayaMalini, K. (2019). Diabetes prediction using different machine learning approaches. 2019 3rd International Conference on Computing and Communications Technologies (IC CCT). IEEE. Retrieved from ieeexplore.ieee.org.
- <https://ieeexplore.ieee.org/abstract/document/8819841>

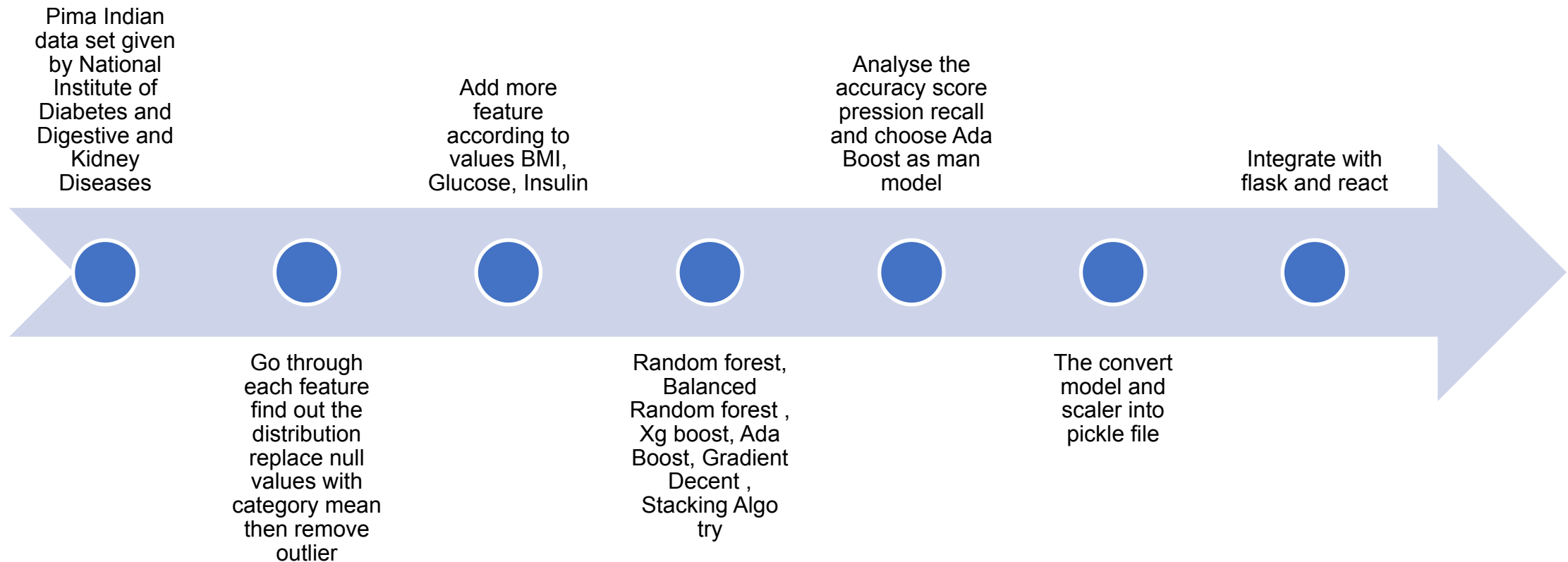


Proposed Design of product....



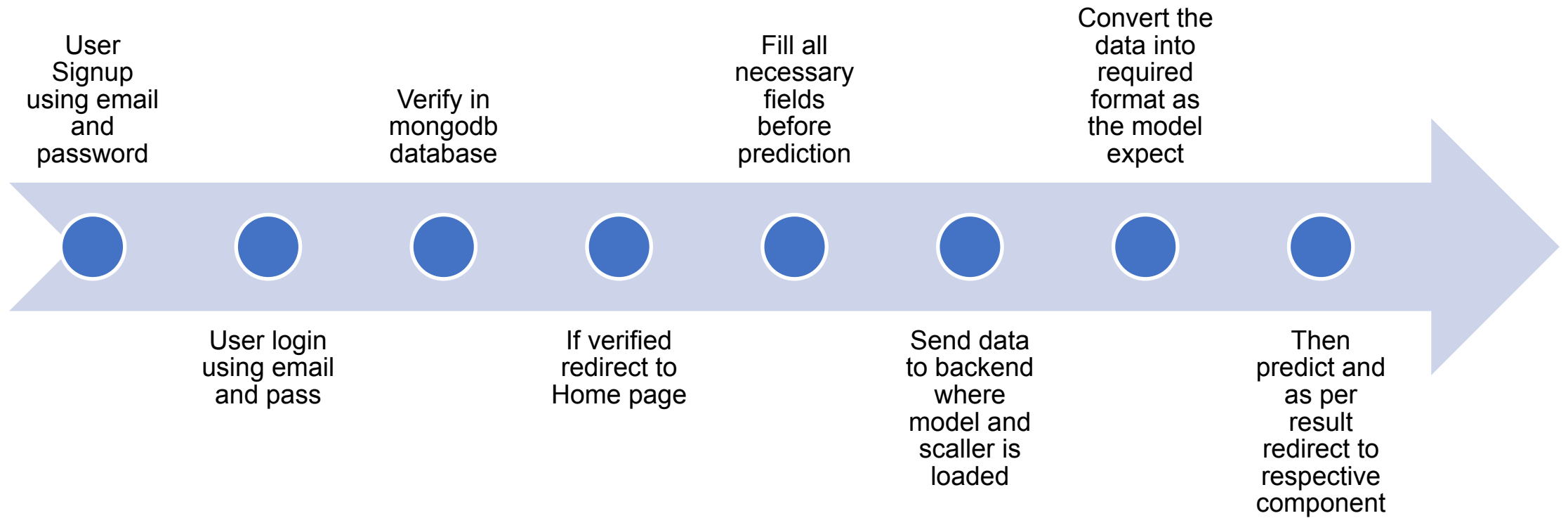


Implementation of Proposed Design of product...





Result Analysis





Result Analysis

- The dataset is provided by the National Institute of Diabetes and Digestive and Kidney Diseases

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1

- Features**
- Pregnancies
- Glucose:- Plasma glucose concentration a 2 hours in an oral glucose tolerance test (140-200)mg/dl
- BloodPressure

- SkinThickness:- represents the triceps skinfold thickness measured in mm
- Insulin: 2-Hour serum insulin microunit/ml (mmu) (16-144)
- BMI:- $\text{Weight kg}/(\text{height in meter})^2$
- Underweight (<18.5), Normal weight (18.5-24.9), Overweight (25-29.9), and Obese (30 or higher). Obesity is further categorized into Class I (30-34.9), Class II (35-39.9), and Class III (40>=)
- DiabetesPedigreeFunction: It is a numeric value that estimates the genetic likelihood of diabetes based on family history (0.1-0.5)



- **Random Forest**:-Random Forest is an ensemble bagging technique where the base models are decision trees. In this approach, sampling is done at the data level — training data is randomly sampled with replacement (bootstrap sampling) to create different subsets for each tree. Additionally, at each node of a decision tree, a random subset of features is considered for splitting (feature randomness). This combination of data randomness and feature randomness helps decorrelate the individual trees and improves the overall performance of the ensemble.
- **AdaBoost** :- AdaBoost is an ensemble boosting algorithm where the base models are usually decision stumps (i.e., single-level decision trees). It works by training the base models sequentially, where each model focuses on correcting the mistakes made by its predecessors.

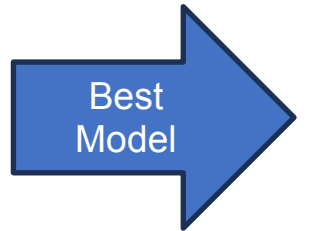


- **Gradient Boosting:-** Gradient Boosting is an ensemble boosting technique that builds a strong predictive model by sequentially combining multiple weak learners, usually decision trees, to minimize a loss function (such as mean squared error or log loss).
- **XG Boost:-** XGBoost stands for Extreme Gradient Boosting. It's a highly optimized implementation of gradient boosting designed for speed and performance
- **Balanced Random Forest:-** A Balanced Random Forest is an adaptation of the classic Random Forest that addresses class imbalance (e.g. 90% class 0 and 10% class 1) in classification problems. Instead of letting the model be biased toward the majority class, Balanced Random Forest ensures each tree in the forest is trained on a balanced bootstrap sample.



Result Analysis

Algo	Accuracy	Class	Precision	Recall	F1-Score
Random Forest	89%	0	0.93	0.94	0.93
		1	0.87	0.85	0.86
XG Boost	91%	0	0.90	0.96	0.93
		1	0.91	0.80	0.85
Ada Boost	95%	0	0.96	0.96	0.96
		1	0.93	0.93	0.93
Gradient Boost	94%	0	0.96	0.95	0.96
		1	0.90	0.93	0.91
Stacking	92%	0	0.93	0.96	0.94
		1	0.92	0.85	0.88
Balanced Random Forest	91%	0	0.96	0.90	0.93
		1	0.82	0.93	0.87





- Use React JS for create single page application
- User give email and name for registration
- Then give email and password for login
- Then go for the home page
- According the prediction result redirect to diabetic or non diabetic page



Result Analysis

Create Account

Full Name

Email address

We'll never share your email with anyone else.

Password

Already have an account? [Login](#)

Sign Up

Login



Welcome Back

Email address

Password

Login

Don't have an account? [Sign up](#)



Diabetes Prediction

Enter your details to check your risk

Number of Pregnancies

Glucose Level

Blood Pressure

Skin Thickness

Insulin Level

Body Mass Index (BMI)

Diabetes Pedigree Function

Age

Predict



Result Analysis

Result

No Worries!!! You don't have Diabetes.



Result

Chances of having Diabetes is more, please consult a Doctor.





Conclusions

- The model provides accurate and timely predictions, improving upon traditional diagnostic methods. People will easily access it with user friendly user interface
- A user-friendly web interface makes the system easily accessible to a wide range of users.
- Early and correct detection helps prevent severe complications by encouraging early medical consultation
- The system empowers users to take proactive steps toward managing or preventing diabetes.
- Overall, the project demonstrates the potential of AI and machine learning in enhancing preventive healthcare and making it more efficient and accessible.



Future Scope

- Accuracy can be improved using the Stacking method with a multi-layer architecture.
- A RAG-based chatbot can be integrated to help users get relevant and reliable information.
- A notification system can also be added to remind users of their daily diet and exercise plans based on their medical condition.
- Can earn money by providing these services to large group of people



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Thank You!